

MongoDB Library Exercise

1. Create Operation

```
test> use libraryDB
switched to db libraryDB
libraryDB> db.createCollection("books")
{ ok: 1 }
libraryDB> db.books.insertOne({
  |   title: "The Alchemist",
  |   author: "Paulo Coelho",
  |   category: "Fiction",
  |   price: 499,
  |   stock: 10,
  |   status: "Available"
  | })
{
  acknowledged: true,
  insertedId: ObjectId('69f761623ed11b6badabc114')
}
libraryDB> db.books.insertMany([
  | {
  |   title: "Atomic Habits",
  |   author: "James Clear",
  |   category: "Self Help",
  |   price: 650,
  |   stock: 15,
  |   status: "Available"
  | },
  | {
  |   title: "Rich Dad Poor Dad",
  |   author: "Robert Kiyosaki",
  |   category: "Finance",
  |   price: 550,
  |   stock: 8,
  |   status: "Available"
  | },
  | ])
```

```
{
  title: "Harry Potter",
  author: "J.K. Rowling",
  category: "Fantasy",
  price: 799,
  stock: 5,
  status: "Out of Stock"
}
])

acknowledged: true,
insertedIds: {
  '0': ObjectId('69f7616c3ed11b6badabc115'),
  '1': ObjectId('69f7616c3ed11b6badabc116'),
  '2': ObjectId('69f7616c3ed11b6badabc117')
}
```

2. Read Operation

```
libraryDB> db.books.find().pretty()
[
  {
    _id: ObjectId('69f761623ed11b6badabc114'),
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    category: 'Fiction',
    price: 499,
    stock: 10,
    status: 'Available'
  },
  {
    _id: ObjectId('69f7616c3ed11b6badabc115'),
    title: 'Atomic Habits',
    author: 'James Clear',
    category: 'Self Help',
    price: 700,
    stock: 15,
    status: 'Available'
  },
  {
    _id: ObjectId('69f7616c3ed11b6badabc116'),
    title: 'Rich Dad Poor Dad',
    author: 'Robert Kiyosaki',
    category: 'Finance',
    price: 550,
    stock: 8,
    status: 'Available'
  },
  {
    _id: ObjectId('69f766bf57156c304ff37481'),
    title: 'Atomic Habits',
    author: 'James Clear',
    category: 'Self Help',
    price: 650,
    stock: 15,
    status: 'Available',
    __v: 0
  }
]
```

```
{
  _id: ObjectId('69f766bf57156c304ff37482'),
  title: 'Rich Dad Poor Dad',
  author: 'Robert Kiyosaki',
  category: 'Finance',
  price: 550,
  stock: 8,
  status: 'Available',
  __v: 0
},
{
  _id: ObjectId('69f766bf57156c304ff37483'),
  title: 'Harry Potter',
  author: 'J.K. Rowling',
  category: 'Fantasy',
  price: 799,
  stock: 5,
  status: 'Out of Stock',
  __v: 0
}
```

3. Update Operation

```
libraryDB>
(To exit, press Ctrl+C again or Ctrl+D or type .exit)
libraryDB> db.books.updateMany(
|   { price: { $lt: 600 } },
|   { $set: { price: 650 } }
| )
|
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 3,
|   modifiedCount: 3,
|   upsertedCount: 0
| }
libraryDB> db.books.updateOne(
|   { title: "Atomic Habits" },
|   { $set: { price: 700 } }
| )
|
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 1,
|   modifiedCount: 1,
|   upsertedCount: 0
| }
libraryDB> db.books.updateOne(
|   { title: "Harry Potter" },
|   { $set: { status: "Available" } }
| )
|
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 1,
|   modifiedCount: 1,
|   upsertedCount: 0
| }
```

4. Delete Operation

```
libraryDB> db.books.deleteMany({ stock: { $lt: 6 } })
{ acknowledged: true, deletedCount: 1 }
libraryDB> db.books.find().pretty()
[
  {
    _id: ObjectId('69f761623ed11b6badabc114'),
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    category: 'Fiction',
    price: 499,
    stock: 10,
    status: 'Available'
  },
  {
    _id: ObjectId('69f7616c3ed11b6badabc115'),
    title: 'Atomic Habits',
    author: 'James Clear',
    category: 'Self Help',
    price: 700,
    stock: 15,
    status: 'Available'
  },
  {
    _id: ObjectId('69f7616c3ed11b6badabc116'),
    title: 'Rich Dad Poor Dad',
    author: 'Robert Kiyosaki',
    category: 'Finance',
    price: 550,
    stock: 8,
```

5. Advanced Query

```
libraryDB> db.books.deleteMany({ available: false })
{ acknowledged: true, deletedCount: 0 }
libraryDB> db.books.deleteOne({ title: "Java Mastery" })
{ acknowledged: true, deletedCount: 0 }
libraryDB> db.books.deleteOne({ title: "Harry Potter"})
{ acknowledged: true, deletedCount: 1 }
libraryDB>
```

```
libraryDB> db.books.find()
[
  {
    _id: ObjectId('69f761623ed11b6badabc114'),
    title: 'The Alchemist',
    author: 'Paulo Coelho',
    category: 'Fiction',
    price: 499,
    stock: 10,
    status: 'Available'
  },
  {
    _id: ObjectId('69f7616c3ed11b6badabc115'),
    title: 'Atomic Habits',
    author: 'James Clear',
    category: 'Self Help',
    price: 650,
    stock: 15,
    status: 'Available'
  },
  {
    _id: ObjectId('69f7616c3ed11b6badabc116'),
    title: 'Rich Dad Poor Dad',
    author: 'Robert Kiyosaki',
    category: 'Finance',
    price: 550,
    stock: 8,
    status: 'Available'
  },
  {
    _id: ObjectId('69f7616c3ed11b6badabc117'),
    title: 'Harry Potter',
    author: 'J.K. Rowling',
    category: 'Fantasy',
    price: 799,
    stock: 5,
    status: 'Out of Stock'
  }
]
```

```
libraryDB> db.books.find({ category: "Finance" })
[
  {
    _id: ObjectId('69f7616c3ed11b6badabc116'),
    title: 'Rich Dad Poor Dad',
    author: 'Robert Kiyosaki',
    category: 'Finance',
    price: 550,
    stock: 8,
    status: 'Available'
  }
]
libraryDB> db.books.find({ price: { $gt: 500 } })
[
  {
    _id: ObjectId('69f7616c3ed11b6badabc115'),
    title: 'Atomic Habits',
    author: 'James Clear',
    category: 'Self Help',
    price: 650,
    stock: 15,
    status: 'Available'
  },
  {
    _id: ObjectId('69f7616c3ed11b6badabc116'),
    title: 'Rich Dad Poor Dad',
    author: 'Robert Kiyosaki',
    category: 'Finance',
    price: 550,
    stock: 8,
    status: 'Available'
  }
]
```


Conclusion:

This exercise demonstrates MongoDB CRUD operations including create, read, update, and delete, along with basic query operators. All operations are performed using mongosh to manage and filter data effectively.